

**BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
HYDERABAD CAMPUS**

Second Semester 2023-2024 Mid-Semester Examinations

BIFS F464: Machine Learning

Dt: 14/03/2024, Type: Closed Book, Weightage: 30%, Duration: 1.5 hrs

Note: There are 4 questions and 4 pages in the question paper. Write your answers **serially** and legibly. All the answers should be written in the answer sheet provided. Calculators (scientific and non-programmable) are allowed. Assume suitable data, if necessary.

Q.1 a) In a multivariate decision tree, very likely, at each internal node, we might not need all the input variables. How can we decrease dimensionality at a node? Describe any two techniques you have studied so far.

[3 Marks]

b) Assume that you wanted to evaluate your chances of getting an A grade in the Midsem grading of Machine Learning course before you appear for the Midsem Exam. Towards this, you want to build up a decision tree based on the following classification. Answer the following:

Studied Hard (SH)	Hours Slept Before the Midsem Exam day (HS)	Lunch Type (LT)	Got A (GA)
No	5	NonVeg	No
No	9	NonVeg	No
Yes	6	NonVeg	No
No	6	Mixed	No
Yes	9	Mixed	Yes
Yes	8	NonVeg	Yes
Yes	8	Veg	Yes
Yes	6	Veg	Yes

- What is the initial entropy of Got A?
- What attribute would the decision tree building algorithm choose for the root of the tree? Justify.
- Draw the full decision tree that would be learned for the above data.

[9 Marks]

c) As we discussed in the class, while boosting and bagging share the goal of improving predictive accuracy, they do so in different ways and are suited for different types of data and models. Describe which one (boosting or bagging) is better for the below scenarios and why?

- Highly imbalanced datasets
- Noisy datasets
- High variance models

[3 Marks]

Q.2 a) For the below two tasks, which model would you choose to build and why (Linear regression or Polynomial regression)? Assume that factors like interpretability, computational resources etc. are not a concern while deciding.

- Predict tomorrow's stock market price given current market conditions and other possible side information.
- Predict the age of a viewer watching a given video type on YouTube.

[3 Marks]

b) Given below is 4 day's class attendance of a course, if the teacher decides to evaluate linear regression models to predict the attendance using the Least-square method (loss function) as discussed in the class, which one he/ she should choose (out of the three given) and why? Reason out.

(P.T.O)

Days (x)	Attendance (y)
1	20
2	10
3	30
4	60

Model 1: $y = 10 + 10x$

Model 2: $y = 10 + 20x$

Model 3: $y = 10 + 5x$

[3 Marks]

c) What is the role of Partial derivatives in the Gradient descent optimization that you studied in the course? How is it related to linear regression? For the below function:

$$f(x_1, x_2, x_3, x_4) = \{[(\cos(x_1x_4) \cdot \sin(x_2^5))/(x_4^2 + (1+x_2^2)/(x_1x_2x_4))]\} + 2x_1^2x_3 - 5x_2x_3$$

Find out:

$$\frac{\partial f}{\partial x_3}(a, b, c, d)$$

[3 Marks]

d) Converting a regression problem into a classification problem involves transforming the target variable from a continuous numerical value to discrete classes. A logistic function is the core function used within the logistic regression algorithm to model the relationship between the features and the binary target variable, as discussed in the class. Describe the logistic function you studied both mathematically and visually. How do you decide a better value for the threshold? Explain any one approach.

[3 Marks]

e) If you use Mean-Squared-Error (MSE) as a loss function for Logistic regression, what problems it might cause to the optimization algorithm? How do you solve this problem? Explain.

[3 Marks]

Q.3 a) Assume that you want to learn a target concept using the version space algorithm studied in the course. Also, assume that the target concept to be learnt is for classifying fruits as either Apples or Non-apples based on two attributes: color and shape. Both attributes have two possible values: Red or Not red for color, and Round or Not round for shape. You are given with three examples as below:

Fruit 1: Red, Round – Apple

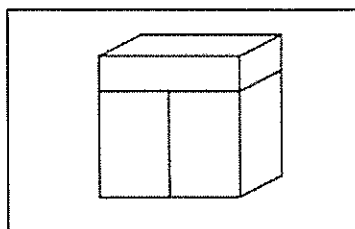
Fruit 2: Not red, Round – Non apple

Fruit 3: Red, Not round – Non apple

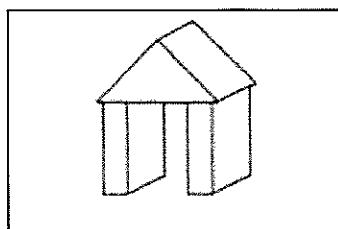
Use Candidate-elimination algorithm discussed to learn the concept. Show clearly the entire hypothesis space with Generic and Specific sets. Iterate through the examples and update the hypotheses.

[4 Marks]

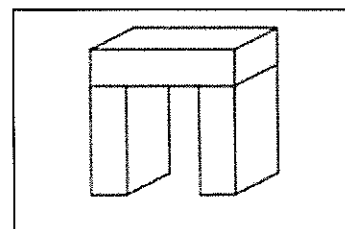
b) Assume that you want to learn a concept Arch (structural description) as discussed in the class. Given three examples to the learner (-ve, +ve, and +ve) in that order, describe how the inductive learner will learn the concept. Use a semantic net representation for the learning. Why is it called hill-climbing approach? Does the sequence of examples play any role in learning (Ex1, Ex2, Ex3)? Explain.



(-ve: Example 1)



(+ve: Example 2)



(+ve: Example 3)

(P.T.O)

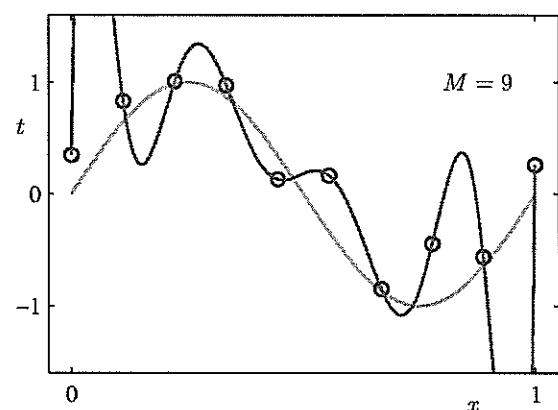
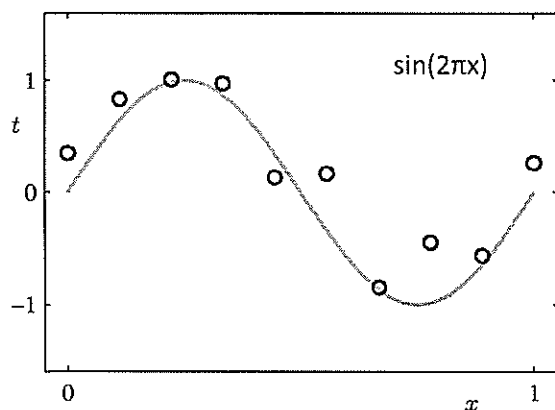
[4 Marks]

c) Given below are three different tasks. Your task is to describe what types of ML algorithms (Supervised, Un-supervised or Reinforcement) you would suggest the learner to use and how i.e., describe these approaches in little detail.

- **Predicting House Prices:** In this task, suppose you have a dataset containing information about houses such as their size, number of bedrooms, location, etc., along with their corresponding sale prices. The task is to predict the price of a house given its features.
- **Customer Segmentation:** In this task, suppose you have a dataset containing information about customers such as their purchase history, demographics, etc., but without any labels. The task is to group similar customers together based on their characteristics.
- **Training a virtual Robot to walk:** In this task, imagine you have a virtual robot in a simulated environment, and the goal is to teach it how to walk. The robot needs to learn which actions to take in different situations to move forward efficiently without falling.

[3 Marks]

d) In the polynomial curve fitting example discussed in the class the below target function (shown left) overfits when $M = 9$ as shown in the right.



The target function and error/ loss functions are as shown below:

$$y(x, \mathbf{w}) = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M = \sum_{j=0}^M w_jx^j \quad E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \{y(x_n, \mathbf{w}) - t_n\}^2$$

How do you use Regularization to control this overfitting? Explain in brief.

[4 marks]

Q.4 a) Amongst many tasks, Machine Learning can help engineers to develop personalized experiences for users by analysing their behaviour and preferences. Engineers can develop ML models to recommend products, customize user interfaces, and tailor content to individual users, enhancing user satisfaction and engagement.

As an example, consider a streaming service like Netflix. Netflix collects vast amounts of data about its users, including their viewing history, ratings, preferences, and even the time of day they watch. By leveraging machine learning algorithms, Netflix can provide highly personalized recommendations to each user.

One of the tasks that Netflix does is: Feedback Loop. As users interact with the platform and provide feedback (e.g., by rating movies or clicking on recommendations), Netflix continuously refines its models to improve the accuracy of its recommendations. This creates a feedback loop where the system becomes more personalized and effective over time.

Your task here is to write down **three** different strategies Netflix might adopt for Refining the existing models based on the feedback received. (Hint: One strategy could be adding more features to the models. Write three other strategies except the one given in the hint. Be brief in answering.)

[3 Marks]

(P.T.O)

b) For the below example dataset (10 samples) for binary classification, compute the confusion matrix. Also, find out Accuracy, Precision, Recall and F1-score of the model. What are your observations for these values? Reason out.

Sample	Predicted class	True class
1	Positive	Positive
2	Negative	Negative
3	Positive	Negative
4	Positive	Positive
5	Negative	Positive
6	Positive	Positive
7	Negative	Negative
8	Positive	Positive
9	Negative	Negative
10	Positive	Negative

[4 Marks]

c) In Fisher's linear discriminant analysis studied in the course, the between-class variance (S_B) and the within-class variance (S_W) are used to find a discriminant function that maximizes the separation between the classes while minimizing the scatter within each class. Fisher's criterion as discussed is given as below:

$$J(w) = S_B / S_W = (m_2 - m_1)^2 / (s_1^2 + s_2^2)$$

Suppose you have a dataset with 2 classes, Class A and Class B, and each class has only one feature x . Using the below data, compute the Fisher's criterion:

Class A:

2, 3, 4

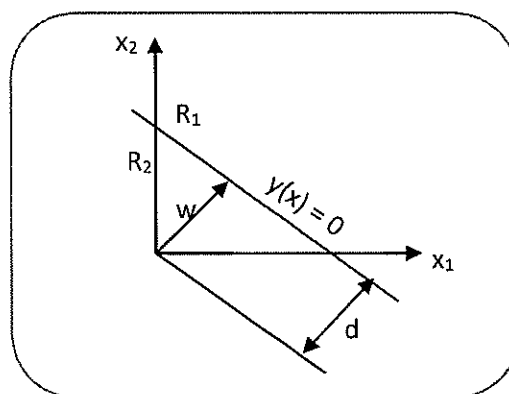
Class B:

6, 7, 8

[4 Marks]

d) The two-class linear discriminant function and its visualization as discussed in the class is given as below:

$$y(x) = w^T x + w_0$$



Show that ' w ' is orthogonal to every vector lying in the decision surface, and hence defines the orientation of the decision surface. Also, find out the distance ' d ' which is the normal distance from origin to the decision surface.

[4 Marks]

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